

Cyclists' mobility and subjective safety in shared urban spaces - a simulator study

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ABSTRACT

The growing adoption of human-centered urban design initiatives in city planning has enhanced attention to cyclists' mobility, safety, and overall travel experience. The effort often results in urban spaces with mixed traffic and it is essential to explore how cyclists behave and perceive their safety in these environments where they more frequently interact with other road users. This study investigates how infrastructure design and traffic volume influence cyclist behavior and subjective safety in a shared paths environment and in unsignalized intersections. Using a cycling simulator, we applied a full factorial design with three two-level (high/low) attributes for each environment: path widths, pedestrian and bicycle flow in shared paths, and vehicle speeds, pedestrian and bicycle flow at unsignalized intersections. Average cycling speed and fixation rate on road users were used as behavioral indicators whereas self-reported perceived unsafety and stress were used as subjective safety indicators. An ordinary least square and ordinal logistic regression model was used to analyze behavioral and subjective safety indicators. Overall, 49 participants completed the experiment. This study quantifies how wider shared paths, higher vehicle speed at unsignalized intersections, and higher pedestrian or bicycle flow in both environments led to lower average cycling speeds and decreased subjective safety. Results show that on shared paths, path width primarily affects cycling speed, while bicycle flow impacts perceived safety. At unsignalized intersections, vehicle speed is the most influential factor for both cycling speed and perceived safety.

1. Introduction

The rising trend of human-oriented urban design initiatives such as 15-minute cities, car-free zones, and low-traffic neighborhoods has increased the focus on cyclists' mobility, safety, and overall riding experience (Buehler & Pucher, 2021; Ballo et al., 2023). In particular, shared urban space has gained attention as an alternative to traditionally segregated street design. Hamilton-Baillie (2008) defines shared space as "a street or place where unnecessary traffic signs, road markings, and signals have been removed and where the different modes of movement — cars, bicycles, pedestrians — interact more directly and on more equal terms." This design philosophy aims to enhance user awareness, negotiation, and safety through reduced formal regulation. Buehler and Pucher (2008) emphasize the importance of integrated, multimodal infrastructure that supports safe and direct interaction among road users — a principle that underpins the shared space approach.

A consequence of this development, however, is that urban cyclists increasingly navigate new types of complex, shared spaces where they must interact with other road users (Mesimäki & Luoma, 2021; Poulos et al., 2017; Takanashi & Murao, 2018). Therefore, there is a need for a better understanding of how such spaces affects cyclists' mobility and subjective safety.

A recent empirical study Chou et al. (2024) found that the prevalence of near-crashes is high in shared paths, and unsignalized intersections. Understanding how path design and traffic flow influence cyclists' behavior and interaction with pedestrians is important to prevent collisions and increase mobility and subjective safety (Zhang et al., 2023, 2017). At unsignalized intersections, cyclists additionally interact with motorized vehicles. This requires heightened awareness and make decision-making more complex due to the reduced predictability of the behavior of the other road users (Mohammadi et al., 2024; Wongwaiwit et al., 2011).

Compared to other environments such as bicycle paths, signalized intersections, and roundabouts, shared paths and unsignalized intersections are studied less frequently. For shared paths and unsignalized intersections, most studies have explored cyclists' critical behaviors and perceived safety using field observations (Che et al., 2021; Huang et al., 2021; Beitel et al., 2018; Boufous et al., 2018; Essa et al., 2018; Mohammadi et al., 2023), and surveys (von Stülpnagel & Rintelen, 2024; Che et al., 2021; Gkekas et al., 2020; Beitel et al., 2018; Ng et al., 2017). However, such methods are not optimal for systematically capturing how external factors such as infrastructure design and traffic conditions influence behavior and subjective safety. Simulator studies, on the other hand, allow systematic variation of such external factors but only a few cycling simulator studies have analyzed shared environments and have either not explored subjective safety (Mohammadi et al., 2024; Zeuwts et al., 2023; Suzuki et al., 2018) or have included subjective safety as part of a broader concept such as mental comfort (Berghoefer & Vollrath, 2023). Through the use of a cycling simulator, this study aims to assess if and how path widths, vehicle speeds, and traffic flow influence cyclists' behavior and subjective safety on

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shared paths and in unsignalized intersections. Specifically, we assess cycling speed, fixation rates, and self-reported perceived unsafety and stress.

The remainder of the paper is structured as follows. Section 2 describes the literature investigating cyclists' behavior and subjective safety using a cycling simulator and literature specifically focusing on shared paths and unsignalized intersections. Section 3 presents the study design, equipment used, data collection, and data analysis. Section 4 includes the main results. Section 5 discusses the results and limitations of the study. Section 6 concludes the paper.

2. Literature review

Table A.8 in Appendix A summarizes studies on cyclists' behavior and subjective safety on shared paths and unsignalized intersections. The table also shows differences in research objectives, data sources, attributes, and indicators of behavior and subjective safety. Key findings are summarized in the following sections.

2.1. Cyclists mobility in shared urban spaces

For shared paths, cyclists ride slower with increasing pedestrian density (Huang et al., 2021; Beitel et al., 2018; Boufous et al., 2018; Essa et al., 2018). Narrower paths were found to reduce cycling speeds based on video observations by Boufous et al. (2018). Beitel et al. (2018) also found higher pedestrian density to increase conflict rates between pedestrians and cyclists. For unsignalized intersections, vehicle speeds, head movements, and fixation influence the cyclists' yielding behavior (Mohammadi et al., 2023, 2024).

2.2. Cyclists comfort and perception of safety in shared urban spaces

Shared paths are designed for multimodal use by pedestrians and cyclists, leading to potential safety concerns due to each group's different speeds and movement patterns (Zhang et al., 2023). In particular, Kazemzadeh et al. (2024) found that the individual level of comfort varies across user types; past experiences, and trip purpose. For example, female cyclists reported feeling notably less comfortable in shared-space interactions than male cyclists. Furthermore, Berghoefer and Vollrath (2023) found that the perceived comfort of cyclists decreases as pedestrian flow increases in shared environments and narrower paths make cyclists feel more unsafe according to a survey by Gkekas et al. (2020). Public support for share spaces can be increased with education and better design Nikiforiadis and Basbas (2019) as well as path segregation and clear signage Batista et al. (2023).

2.3. Behavioral observation and attention

Some studies have more specifically analyzed cyclists' and pedestrians' interaction patterns. While cyclists (and scooter users) pay more attention to other road users' movement direction and interactions compared to pedestrians Pashkevich et al. (2022), their yielding behavior, spacing, and speed adjustments vary with traffic density and user type. Stülpnagel (2020) found that cyclists in open, complex environments distributed their gaze across a wider range of angles rather than focusing straight ahead and that this effect decreases with more experience and increases with higher level of perceived risk. Mohammadi et al. (2024) found that cyclists focus more on vehicles due to uncertainty about whether the vehicles will yield to them.

2.4. Study methods

Most previous studies rely on surveys or field (video) observations, limiting the ability to control specific attributes. Cycling simulators on the other hand, can create realistic traffic conditions in controlled environments, offering researchers precise control over the experimental design without exposing participants to real-world dangers (Cobb et al., 2021; O'Hern et al., 2017; Meuleners & Fraser, 2015).

Most studies using cycling simulators investigate environments where cyclists' behavior is regulated by clear instructions through infrastructure design such as roundabouts (Friel et al., 2023; Zeuwts et al., 2023; Xu et al., 2023), bicycle paths (Mohammadi et al., 2024; Guo et al., 2023; Meuleners et al., 2023; Nazemi et al., 2021; Bogacz et al., 2020), and traffic lights (Friel et al., 2023; Scott-Deeter et al., 2023; Zeuwts et al., 2023; Xu et al., 2023; Cobb et al., 2021; Kwigizile et al., 2017; Liu et al., 2012). A few studies have conducted controlled experiments for shared paths and unsignalized intersections, either using a cycling simulator (Mohammadi et al., 2024; Berghoefer & Vollrath, 2023), or a closed-field experiment (Huang et al., 2021). Mohammadi et al. (2024) focused solely on cyclists' yielding behavior without considering perceived safety while Berghoefer and Vollrath (2023) examined infrastructure preferences, and Huang et al. (2021) studied interactions only within a campus setting, where cyclist and pedestrian movements were regulated in a closed circular route.



Fig. 1. The cycling simulator and eye-tracker used in the study.

2.5. Research gap

A comprehensive and systematic understanding and quantification of how variations in infrastructure design and traffic flow affect cyclists' mobility and perceived safety at shared paths and unsignalized intersections remains unexplored. Due to the limited detail and quantitative understanding of how cyclists behave and perceive safety in shared urban spaces, this study aims to quantify how path width and pedestrian/bicycle flow on shared paths, and vehicle speed and pedestrian/bicycle flow at unsignalized intersections impact cyclists' mobility and perceived safety using a cycling simulator. Based on results from existing literature regarding different types of infrastructure we anticipate that narrower shared paths, could lead to lower average riding speed (Boufous et al., 2018) and higher perceived unsafety (Gkekas et al., 2020). For the unsignalized intersections with higher vehicle speeds, we anticipate that cyclists might choose a lower riding speed (Mohammadi et al., 2023, 2024), pay more attention to the vehicles (Mohammadi et al., 2024), and feel less safe. In both environments, we expect that increasing pedestrian/cyclist flow to leads to reduced average riding speeds and cyclists feeling less safe (Huang et al., 2021; Beitel et al., 2018; Boufous et al., 2018; Essa et al., 2018).

3. Methodology

This section describes the technical setup of the cycling simulator, how the experiment is designed and implemented, and how the data is collected and processed.

3.1. Technical setup

The cycling simulator utilized in this research (shown in Fig. 1) was developed by the Traffic Engineering and Control Department at the Technical University of Munich, Germany. The setup includes a stationary bike trainer with a bicycle mounted on it, positioned in front of three 55-inch screens. One screen is directly in front of the bicycle, while the other two are placed at an angle of approximately 74 degrees on either side. The road layouts and built environment were created in RoadRunner (RoadRunner, 2024), and the flow and movement of all road users were created in Unity. For more hardware and software setup details see Keler et al. (2018).

To measure the participants-fixation on different objects an eye-tracker from Pupil Core (Pupil Core, 2024) is used. In this study, fixation is defined when the maximal distance between all gaze locations is within 1.5 degrees of visual angle and occurs within a duration window between 90 to 500 milliseconds, following the criteria set by Negi and Mitra (2020) and Velichkovsky et al. (2002).

3.2. Experimental design

The experiment consists of two environments; a shared path and an unsignalized intersection, both shown in Fig. 2. Their geometries are shown in Fig. 3. In the shared path environment, pedestrians and cyclists are expected to share the full width of the road without any priority or dedicated areas to either pedestrians or cyclists. Even though some deviations to speed and direction

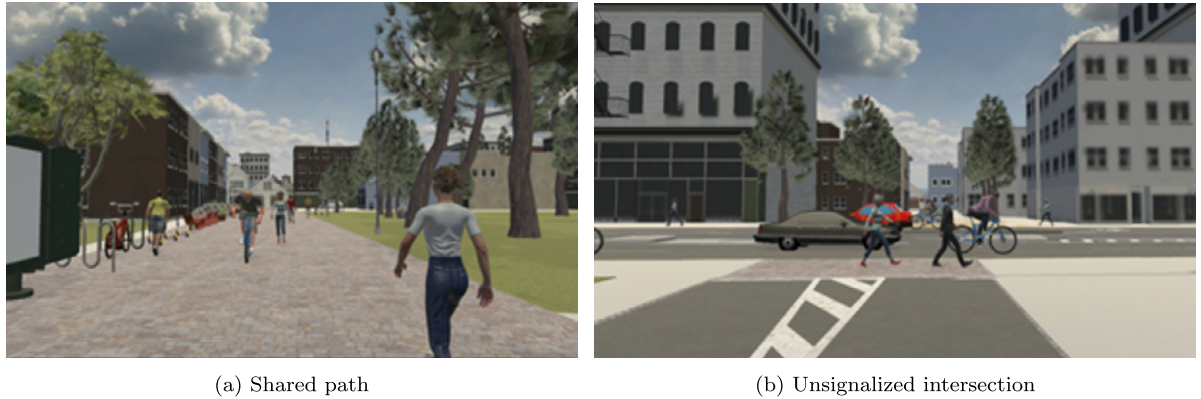


Fig. 2. Infrastructure in cycling simulator environment.

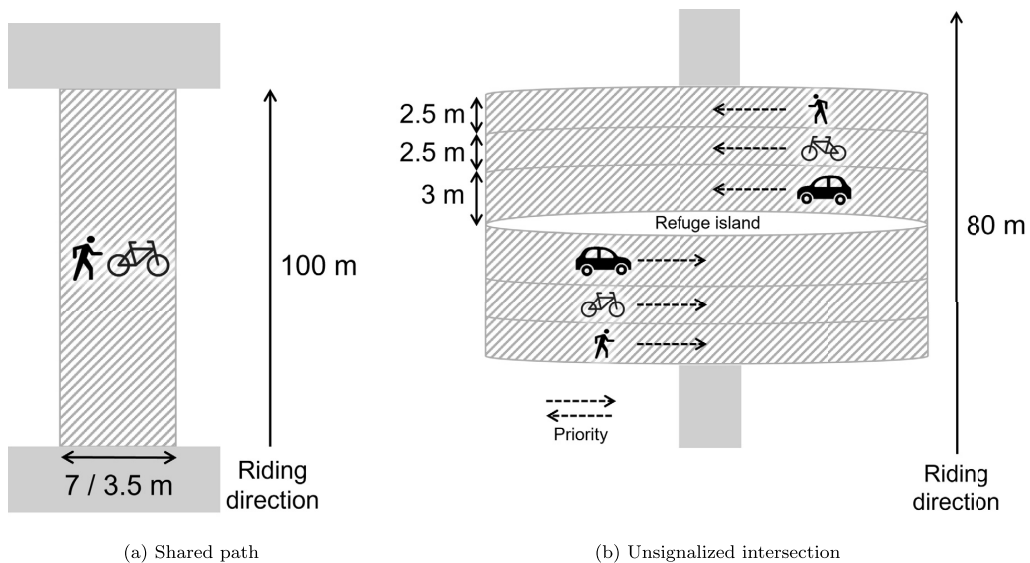


Fig. 3. Geometry of infrastructure environment.

could be expected in real life, we assume that most users of the shared path maintain their speed and direction to reduce complexity in the setup of the experiment. At the unsignalized intersection the road users on the minor road (including the participants) must give way to the crossing traffic on the primary road. Therefore, in the experiment, all road users on the primary road are programmed to maintain their speed and direction without yielding to the participant. The software did not allow for any response in case of a collision with other road users.

To understand how traffic and road conditions affect cyclist behavior and subjective safety, we employ a full factorial design containing three attributes with two levels for each environment, resulting in eight unique combinations of attribute levels per environment (see Table 1). We define a scenario as a unique combination of an attribute level and an environment. Furthermore, we define a run as a pairing containing one shared path scenario and one unsignalized intersection scenario. Eight of such runs (labeled 1–8) are constructed. To reduce the learning effect, we swap the order of the environments in four of the runs. Participants start with the shared path, followed by the unsignalized intersection in runs 1, 2, 5, and 7, whereas the order is reversed for the other four runs. As an example, Run 4 includes an unsignalized intersection scenario characterized by high vehicle speed and low pedestrian and bicycle flow (H/L/L), followed by a shared path scenario with high path width, high pedestrian flow, and low bicycle flow (H/H/L).

Each participant completed eight runs. To minimize order effects and ensure that participants did not ride in the same sequence from run 1 to 8, the order of the runs was randomized into eight distinct courses using a Balanced Latin Square Generator (Damien Masson, 2024). The total duration of the eight runs was approximately 20 minutes.

Table 1
Experimental design.

Environment	Attribute	Levels Low (L) / High (H)	Run							
			1	2	3 ^a	4 ^a	5	6 ^a	7	8 ^a
Shared path	Path width	3.5 / 7 [meters]	H	L	L	H	L	H	H	L
	Pedestrian flow	250 / 500 [pedestrians/hour]	H	L	H	H	L	L	L	H
	Bicycle flow	250 / 500 [cyclists/hour]	H	L	L	L	H	H	L	H
Unsignalized intersection	Vehicle speed	20 / 40 [km/hour]	L	L	H	H	H	L	L	H
	Pedestrian flow	500 / 1,000 [pedestrians/hour]	L	H	H	L	H	H	L	L
	Bicycle flow	350 / 700 [cyclists/hour]	L	H	H	L	L	L	H	H

^a The order of environment is reversed (i.e., unsignalized intersection before the shared path)

3.2.1. Attribute levels

The path width attributes are selected based on road design guidelines from the Cycling Embassy of Denmark (Andersen et al., 2012) and the German Research Association for Road and Traffic Engineering (FGSV, 2010), which recommend a minimum width wider than 3 meters. Thus, we use a path width of 3.5 meters as the narrow attribute level. To assess how a wider path impacts cyclists' behavior and psychological factors, we also include a 7-meter path, which is also common in many European cities. In a cycling simulator environment replicating the German context, Berghoefer and Vollrath (2023); Nazemi et al. (2021) chose 3.6 and 3.5 meters as the path width, respectively.

The pedestrian flows are defined according to Cyclenation (2014) and Skoczyński (2021), which suggest that the maximum pedestrian flow on a shared path should be approximately 100 pedestrians per hour per meter of width. Thus, we use 500 pedestrians per hour as the higher attribute level which is equivalent to 71 and 142 pedestrians per hour per meter for wide and narrow paths, respectively. To compare with a lower pedestrian flow and explore how the amount of pedestrian flow influences cyclists' behavioral and psychological factors, we selected 250 pedestrians per hour as the lower attribute level. For the attribute levels of bicycle flow, we selected 250 and 500 cyclists per hour as low and high levels, respectively. These represent reasonable flow rates on a shared path and are distinct enough for meaningful comparison.

For the selection of attribute levels for the three attributes at the unsignalized intersection, there are no strict criteria for vehicle speed and pedestrian and bicycle flow (Andersen et al., 2012; Jensen, 2012). Therefore, the attribute levels were chosen by referring to the unsignalized intersection with one of the highest near-crash-rate locations identified in Chou et al. (2024) in Metropolitan Copenhagen, Denmark. We use 40 km/h and 20 km/h for levels of high and low vehicle speed. Although the allowed speed at this location is 50 km/h (similar to most urban roads in most EU countries (Commission, 2024)), actual driving speeds are usually lower as drivers anticipate crossing road users at the intersection and expect nearby traffic signals. Therefore, we chose 40 km/h as the higher level for vehicle speed. We conducted a pilot test to examine whether the lower level for vehicle speed should be 30 km/h or 20 km/h, and eventually chose 20 km/h as participants perceived a clearer distinction from 40 km/h at this speed. In the experiment, scenarios with vehicle speeds of 40 km/h and 20 km/h feature the same number of vehicles passing through the unsignalized intersection, leading to vehicle flows of 500 and 1,000 vehicles per hour, respectively. The experiment also ensures that participants have sufficient gaps to cross the unsignalized intersection in both vehicle-speed scenarios. For pedestrian and bicycle flow, we selected 500 and 1,000 pedestrians per hour and 350 and 700 cyclists per hour as the high and low attribute levels, ensuring they are reasonable and distinguishable.

3.3. Indicators

All behavioral and subjective safety indicators measured during the experiment are shown in Table 2. In both infrastructure environments, the participant's average speed is calculated by dividing the distance traveled by the participant by the total crossing time, determined from the start and end coordinates of infrastructure along with the corresponding time stamps. At unsignalized intersections, the total crossing time includes cyclists approaching the intersection, waiting time before starting to cross, and actual crossing time.

The fixation rate on road users in both environments is measured with an eye-tracker. Fixation events on pedestrians, cyclists, or vehicles are manually annotated. The rate for each road user is calculated by dividing the total fixation events on them by the total crossing time in each environment.

Perceived unsafety and stress were assessed as subjective safety indicators using a 5-point Likert scale in both infrastructure environments. Perceived unsafety ranged from "very safe" to "very unsafe," reflecting participants' subjective sense of security while riding through each scenario. Perceived stress ranged from "no stress" to "very high stress," capturing participants' emotional strain while riding through each scenario. Both indicators were measured through post-run surveys conducted immediately after participants completed each run (one for the shared path and one for the unsignalized intersection).

Table 2
Behavioral and subjective safety indicators.

Environment	Variable	Measurement
Shared path and	Participant average speed [km/h]	Average speed passing through infrastructure
	Fixation rate [%]	Percentage of total fixation duration on each road user to total crossing time
	Perceived unsafety [5-point Likert]	1: Very safe 2: Somewhat safe 3: Neither safe nor unsafe 4: Somewhat unsafe 5: Very unsafe
Unsignalized intersection	Perceived stress [5-point Likert]	1: No stress 2: Low stress 3: Moderate stress 4: High stress 5: Very high stress

3.4. Experimental procedure

The study was carried out in May and June of 2024. 50 participants were recruited from various sources including the online platforms of the Technical University of Munich, the mailing list of the Chair of Traffic Engineering and Control, and the transportation-related institutions in the city of Munich (e.g., Municipality and German National Cyclists' Association Munich branch). During the registration process, participants were asked to complete a pre-study survey covering demographic information, cycling habits, and their desired time for experiment participation. During the experiment, participants were briefed about the study, instructed on cycling simulator usage with a test ride, and asked to provide informed consent before the experimental ride. Once participants had familiarized themselves with the cycling simulator in the test ride, they started the experiment. After each of the eight runs, participants had a short break and were asked to rate their perceived unsafety and stress with a 5-point Likert scale about the shared path and the unsignalized intersection scenarios. The average completion time for the entire experiment was approximately one hour and participants received a 15-euro shopping coupon as a reward. Fig. 4 illustrates the overall methodological framework, outlining the sequential steps of data acquisition, processing, and analysis in the study.

3.5. Statistical analysis

After data was collected and processed, ordinary least square (OLS) regression was used to explore how independent variables affect behavioral indicators, and ordinal logistic regression was used to explore subjective safety indicators.

The specification of the OLS shown in formula (1) presenting the relation between the behavior indicators, *Speed*, which denotes participant average speed, and the participant i in scenario j .

$$\begin{aligned}
 Speed_{i,j} = & \alpha + \beta_1 Path\ width_j + \beta_2 Pedestrian\ flow_j + \beta_3 Bicycle\ flow_j \\
 & + \beta_4 Path\ width_j \times Pedestrian\ flow_j \\
 & + \beta_5 Path\ width_j \times Bicycle\ flow_j \\
 & + \beta_6 Pedestrian\ flow_j \times Bicycle\ flow_j \\
 & + \beta_7 \cdot Individual\ dummies_i + \varepsilon.
 \end{aligned} \tag{1}$$

Here α denotes the intercept of the model that is the expected mean when all explanatory variables are zero. *Path width* denotes the path width at the shared path, while at the unsignalized intersection, this parameter is replaced by *Vehicle speed*. *Pedestrian flow* denotes pedestrian flow at both the Shared path and the unsignalized intersection. Similarly, *Bicycle flow* denotes bicycle flow at both infrastructure environments. All aforementioned variables are dummies which are 1 when the high level is used, and 0 when the low attribute level is used. *Individual dummies* denotes column vectors of dummy variables including individual participants' characteristics. β_1 to β_3 denotes coefficient of explanatory variables. β_4 to β_6 denotes coefficient of interaction terms between two explanatory variables. β_7 denotes vectors including coefficients of individual dummy variables. ε denotes the error term indicating the effects of factors not captured by the explanatory variables.

To examine the influence of behavioral and subjective indicators on fixation patterns toward vehicles at unsignalized intersections,

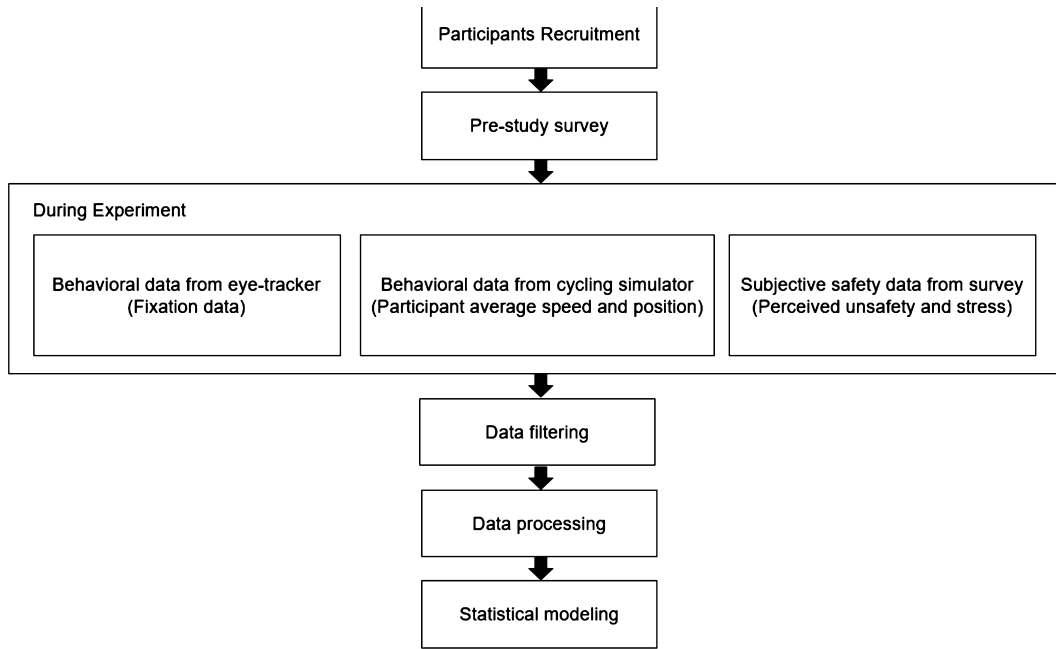


Fig. 4. Data flow chart.

the OLS specification presented in Equation (2) was formulated.

$$\begin{aligned}
 \text{Fixation on vehicles}_{i,j} = & \alpha + \beta_1 \text{Speed}_j + \beta_2 \text{Unsafety} \\
 & + \beta_3 \text{Vehicle speed}_j + \beta_4 \text{Pedestrian flow}_j + \beta_5 \text{Bicycle flow}_j \\
 & + \beta_6 \text{Vehicle speed}_j \times \text{Pedestrian flow}_j \\
 & + \beta_7 \text{Vehicle speed}_j \times \text{Bicycle flow}_j \\
 & + \beta_8 \text{Pedestrian flow}_j \times \text{Bicycle flow}_j \\
 & + \beta_9 \cdot \text{Individual dummies}_i + \varepsilon.
 \end{aligned} \tag{2}$$

In this context, *Fixation on vehicles* refers to the fixation rate on vehicles. It is important to note that *Unsafety* is treated as a continuous numerical variable rather than an ordinal one, as the differences between levels of perceived unsafety are both meaningful and quantifiable. This treatment supports the appropriate application of the OLS estimation.

The ordinal regression model is presented in equation (3) below presenting the relation between the subjective safety indicators, *Unsafety*, which denotes perceived unsafety, and the participant *i* in scenario *j*.

$$\begin{aligned}
 & P(\text{Unsafety} \leq n), \quad n = 1, 2, 3, 4, 5 \\
 \text{Odds} = & \frac{P(\text{Unsafety} \leq n)}{1 - P(\text{Unsafety} \leq n)} = \frac{P(\text{Unsafety} \leq n)}{P(\text{Unsafety} > n)} \\
 \log(\text{Odds}) = & \log\left(\frac{P(\text{Unsafety} \leq n)}{P(\text{Unsafety} > n)}\right) = \text{Logit}(P(\text{Unsafety} \leq n)) \\
 \text{Logit}(P(\text{Unsafety}_{i,j} \leq n)) = & \alpha_n + \beta_1 \text{Speed}_j \\
 & + \beta_2 \text{Fixation rate}_j \\
 & + \beta_3 \text{Path width}_j + \beta_4 \text{Pedestrian flow}_j + \beta_5 \text{Bicycle flow}_j \\
 & + \beta_6 \text{Path width}_j \times \text{Pedestrian flow}_j \\
 & + \beta_7 \text{Path width}_j \times \text{Bicycle flow}_j \\
 & + \beta_8 \text{Pedestrian flow}_j \times \text{Bicycle flow}_j \\
 & + \beta_9 \cdot \text{Individual dummies}_i + \varepsilon.
 \end{aligned} \tag{3}$$

Here *Unsafety* has $n = 5$ categories and $P(\text{Unsafety} \leq n)$ represent the cumulative probability of being in category n or below, and it is modeled across all threshold categories n . Odds represents the odds of being in a lower category versus a higher category of the

Table 3

Model estimation results for participant average speed at the shared path.

Variables	Coefficient	Std. error	t-value	p-value
Participant average speed [km/hour]				
Intercept	17.159	1.185	14.483	<0.001 ***
Path width high	2.832	0.325	8.702	<0.001 ***
Pedestrian flow high	-1.751	0.460	-3.803	0.001 ***
Bicycle flow high	-1.521	0.460	-3.304	0.001 ***
Pedestrian flow high $\times$ Bicycle flow high	1.870	0.651	2.873	0.004 ***
Adjusted R-squared	0.485			

* (p-value <0.05), ** (p-value <0.01), and *** (p-value <0.005).

ordinal outcome. α_n denotes the intercept (also called thresholds or cut points) of the model that distinguishes between the different levels of the ordinal dependent variable.

We included Participant average speed *Speed* as an independent variable to investigate whether behavioral and subjective indicators are interconnected. At the unsignalized intersection, the *Path width* is replaced by *Vehicle speed*. We also included *Fixation* as an independent variable since it is worth investigating the relationship between perceived unsafety and the participant's gazing behavior. The remaining independent variables and coefficients are as same as the OLS model shown in formula (1).

After estimating the models, the odds ratios (ORs) for each explanatory variable were obtained by exponentiating the estimated coefficients (i.e., $e^{\beta_{1,2,\dots,9}}$). ORs provide a clear indication of how a one-unit change in an independent variable influences the associated odds.

3.6. Sample

A total of 50 participants attended the experiment. One participant dropped out due to motion sickness, and was discarded from the data. Participants were selected based on the criteria of cycling at least once per month and having no physical or psychological impairments. The eye movements of four participants could not be detected, likely due to reflections from glasses or makeup. Therefore, eye-tracker fixation data is only available for 45 participants. The 49 participants who completed the experiment included 16 (32.7%) females and 33 (67.3%) males. Their age composition is 20-29 (11, 22.4%), 30-39 (11, 22.4%), 40-49 (12, 24.5%), 50-59 (10, 20.4%), 60-69 (3, 6.1%), and 70-79 (2, 4.1%). For their cycling frequency in a recent year, 30 (61.2%) ride almost every day, 10 (20.4%) ride every two to three days in a week, 5 (10.2%) ride once a week, 1 (2.0%) rides once every two weeks, and 3 (6.1%) ride once a month. For their trip purpose, 25 (51.0%) ride for work, 2 (4.1%) ride for school, 7 (14.3%) ride for leisure, 6 (12.2%) ride for errand, 3 (6.1%) ride for social activities, and 6 (12.2%) ride for other purposes.

4. Results

This section will present the descriptive analysis of the data collected from the experiment and the outcomes of the model estimation results. In the model estimation results, all parameters are significant at the 5% level. In [Appendix C](#) we present the models with all tested attributes. It is important to mention that all models account for individual participants' characteristics as dummy variables (i.e., **Individual dummies**; in formula (1)), but these are not included in the tables since they are not the focus of our analysis.

4.1. Analysis of cyclists in behavior in the shared path scenario

Participant average speed

Fig. 5 shows boxplots of participant average speeds across the eight shared path scenarios (for more information on the scenarios, we refer to Table 1). As seen in the figure, the width of the shared path seems to affect speed, as the four "wide" scenarios also have the highest speed. When both the bicycle and pedestrian flow is low, the participant's average speed is higher than in all other combinations of the flow attributes in the narrow and wide scenarios.

For the model estimating participant average speed on the shared path shown in Table 3, all three attributes and the interaction between pedestrian flow and bicycle flow are significant. The average speed rises the most (by 2.8 km/hour) when the path width expands from narrow (3.5 meters) to wide (7 meters). The average speed decreases by 1.8 km/hour when pedestrian flow rises from low (250 pedestrians/hour) to high (500 pedestrians/hour). Similarly, the average speed decreases by 1.5 km/hour when bicycle flow rises from low (250 cyclists/hour) to high (500 cyclists/hour). When both pedestrian and bicycle flows increase from low to high, the average speed decreases by 1.4 km/hour due to the interaction effect.

Perceived unsafety

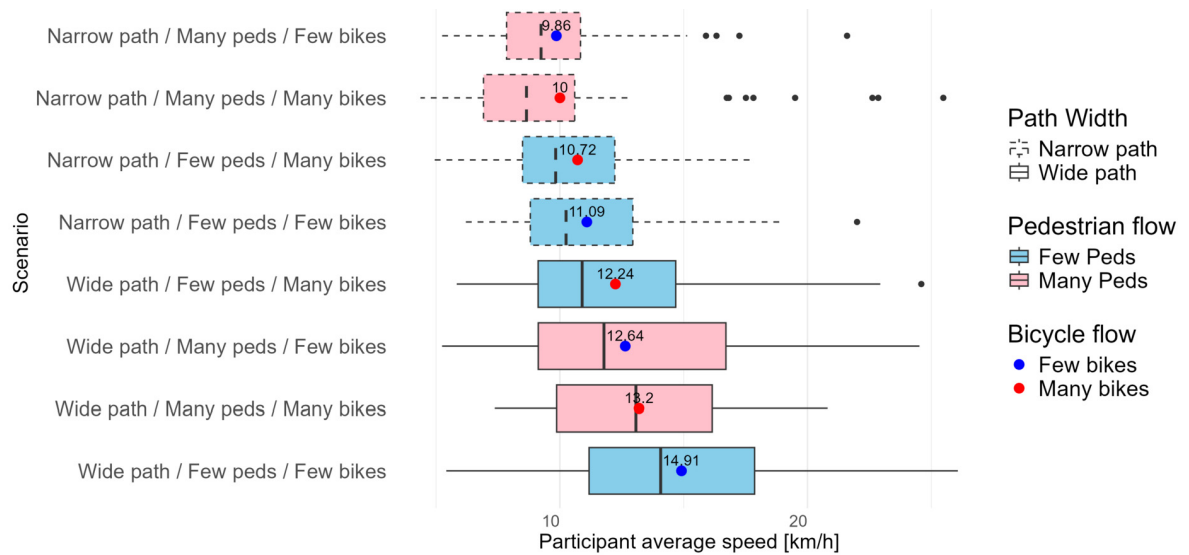


Fig. 5. Participant average speed in eight scenarios (combination of environment and variable values) at the shared path environment. Dots indicate the mean value.

Table 4

Model estimation results for perceived unsafety at the shared path.

Variables	Odds ratio	Std. error	t-value	p-value
Intercepts (Thresholds)				
Perceived unsafety [Likert]				
Perceived unsafety 1 vs 2 (or higher)	0.019	0.549	-7.209	<0.001 ***
Perceived unsafety 2 (or lower) vs 3 (or higher)	0.295	0.502	-2.431	0.015 *
Perceived unsafety 3 (or lower) vs 4 (or higher)	2.368	0.507	1.702	0.089
Perceived unsafety 4 (or lower) vs 5	99.086	0.722	6.363	<0.001 ***
Participant average speed high	0.887	0.033	-3.672	<0.001 ***
Path width high	0.458	0.319	-2.447	0.014 *
Pedestrian flow high	1.728	0.216	2.532	0.011 *
Bicycle flow high	3.145	0.304	3.768	<0.001 ***
Path width high × Bicycle flow high	0.326	0.431	-2.600	0.009 **
Akaike information criterion (AIC)	791.16			

* (p-value <0.05), ** (p-value <0.01), and *** (p-value <0.005).

Fig. 6 shows boxplots of participants indicated subjective safety across the eight shared path scenarios. We only show the results of perceived unsafety since the correlation between perceived unsafety and stress is very high (74% in both environments). The corresponding figures for perceived stress are in Appendix B.

Participants riding in narrow shared-path scenarios have higher perceived unsafety than those riding in wide-shared-path scenarios. Moreover, participants riding in narrow shared-path scenarios felt more unsafe when pedestrian and bicycle flow increased from low to high simultaneously.

For the model estimating perceived unsafety on the shared path shown in Table 4, participant average speed, all three attributes, and interactions between path width and bicycle flow are significant. Participant average speed and path width have odds ratios lower than 1, meaning that faster average speed and wider paths are associated with a lower level of perceived unsafety. The odds of perceiving a higher level of unsafety decrease by about 54.2% (i.e., $1 - 0.458$) when increasing the width from narrow to wide. For comparison, a similar effect can be found when increasing the speed by 6.5 km/h (i.e., $1 - (0.887)^{6.5} = 0.54$).

Pedestrian and bicycle flow have odds ratios larger than 1, meaning that an increase from low to high in both cases is associated with a higher likelihood of perceived unsafety. The odds of high-level perceived unsafety increase by about 72.8% and 214.5%, respectively. When both path width and bicycle flow increase from low to high, the perceived unsafety does not change much compared to low-path-width and high-bicycle-flow scenarios due to the interaction effect. This indicates that in the low path-width scenario, the bicycle flow does not play an important role.

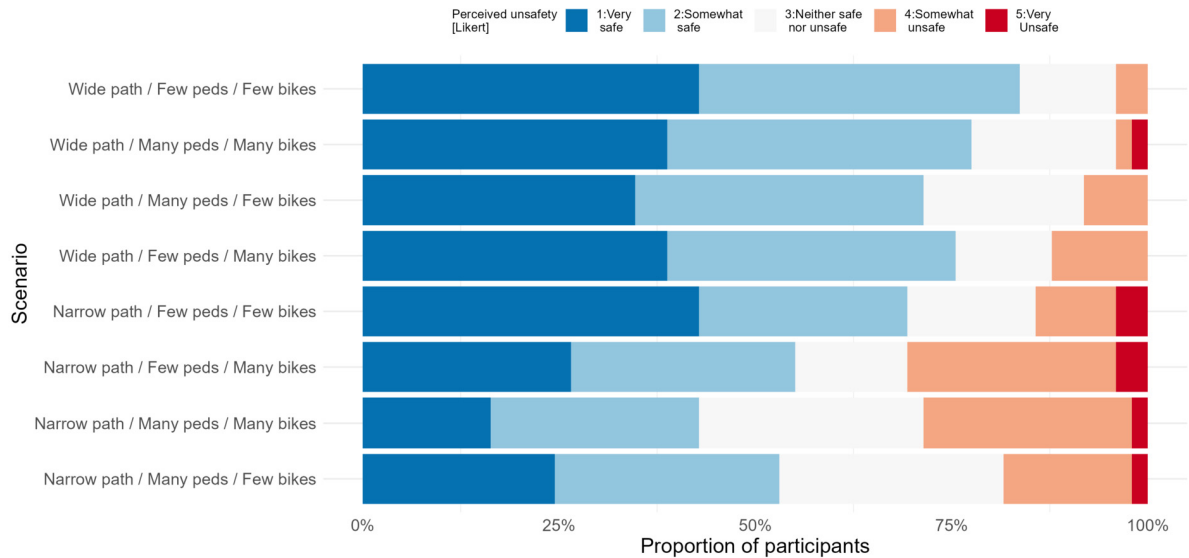


Fig. 6. Perceived unsafety in eight scenarios at the shared path environment.

Table 5

Model estimation results for participant average speed at the unsignalized intersection.

Variables	Coefficient	Std. error	t-value	p-value
Participant average speed [km/hour]				
Intercept	15.378	1.598	9.626	<0.001 ***
Vehicle speed high	-6.678	0.621	-10.752	<0.001 ***
Pedestrian flow high	-3.486	0.621	-5.612	<0.001 ***
Vehicle speed high × Pedestrian flow high	4.547	0.878	5.177	<0.001 ***
Adjusted R-squared	0.390			

* (p-value <0.05), ** (p-value <0.01), and *** (p-value <0.005).

4.2. Unsignalized intersection

Participant average speed

Fig. 7 shows that at unsignalized intersections, participant average speed is higher when the vehicle speed is low compared to when the vehicle's speed is high. Among the slow-vehicle scenarios, the participant average speed is lower when pedestrian flow is high compared to low pedestrian flow.

For the model estimating participant average speed at the unsignalized intersection shown in Table 5, vehicle speed, pedestrian flow, and their interaction are significant. The average speed reduces by 6.7 km/hour when the vehicle speed increases from low (20 km/hour) to high (40 km/hour), and it reduces by 3.5 km/hour when the pedestrian flow increases from low (500 pedestrians/hour) to high (1000 pedestrians/hour). When both vehicle speed and pedestrian flow increase from low to high, the average speed reduces by 5.7 km/hour.

Perceived unsafety

The perceived unsafety at the unsignalized intersection, shown in Fig. 8, indicates that participants had higher perceived unsafety when they rode in high-vehicle-speed scenarios.

For the model estimating perceived unsafety at the unsignalized intersection shown in Table 6, participant average speed, vehicle speed, bicycle flow, and fixation on vehicles are significant. Participant average speed has an odds ratio lower than 1, meaning that for every kilometer per hour increase in average speed, the odds of perceiving higher levels of unsafety decrease by 21.4%. Vehicle speed, bicycle flow, and fixation on vehicles have odds ratios larger than 1, meaning that for every unit increase, the odds of perceiving higher levels of unsafety increase by 264.3%, 82.8%, and 6%, respectively, showcasing that vehicle speed by far is the most influential factor.

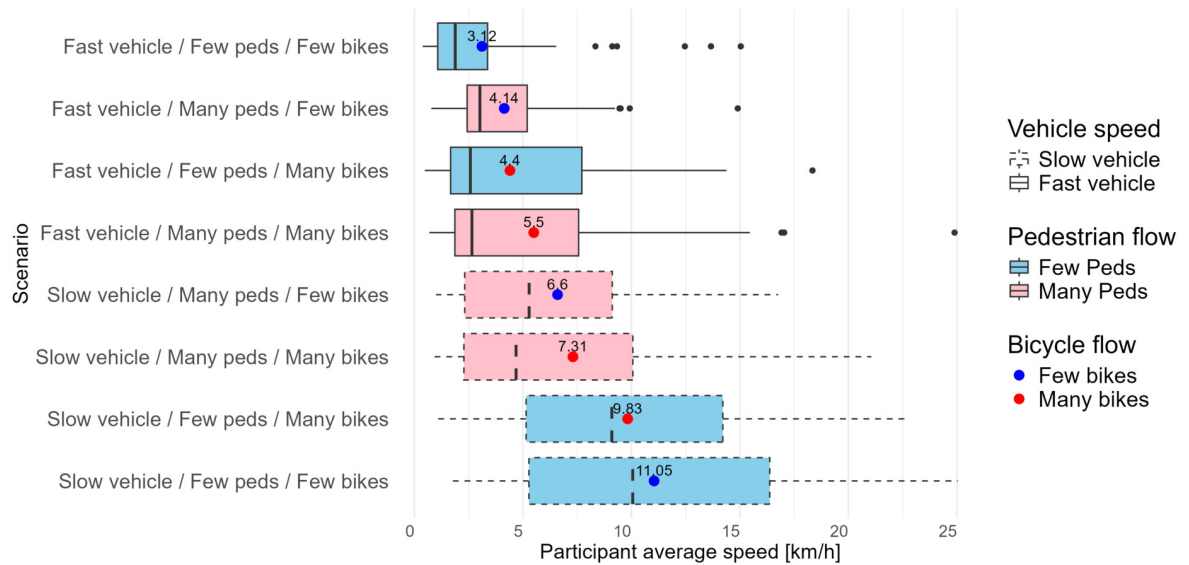


Fig. 7. Participant average speed in eight scenarios at unsignalized intersection environment. Dots indicate the mean value.

Table 6

Model estimation results for perceived unsafety at the unsignalized intersection.

Variables	Odds ratio	Std. error	t-value	p-value
Intercepts (Thresholds)				
Perceived unsafety [Likert]				
Perceived unsafety 1 vs 2 (or higher)	0.004	0.813	-6.886	<0.001 ***
Perceived unsafety 2 (or lower) vs 3 (or higher)	0.063	0.746	-3.707	<0.001 ***
Perceived unsafety 3 (or lower) vs 4 (or higher)	0.453	0.731	-1.084	0.278
Perceived unsafety 4 (or lower) vs 5	3.244	0.736	1.598	0.110
Participant average speed high	0.786	0.030	-8.130	<0.001 ***
Vehicle speed high	3.643	0.251	5.155	<0.001 ***
Bicycle flow high	1.828	0.217	2.775	0.006 **
Fixation on vehicles (%)	1.060	0.019	3.139	0.002 ***
Akaike information criterion (AIC)	831.19			

* (p-value <0.05), ** (p-value <0.01), and *** (p-value <0.005).

Fixation on vehicles

For the model estimating fixation on vehicles at the unsignalized intersection shown in Table 7, participant average speed, vehicle speed, and pedestrian flow are significant. A drop in average participant speed of 1 kilometer per hour is associated with a 0.3% increase in fixation on vehicles. Fixation is also reduced by 2.2% when pedestrian flow increases from low to high. The fixation on vehicles rises by 3.5% when vehicle speed rises from low to high.

5. Discussion

Our findings provide important evidence on how the width of a shared path influences cyclists' perceived safety and riding behavior. In line with Gkekas et al. (2020) and Hummer et al. (2005), we observe that cyclists feel significantly safer when path width is increased. One likely reason is the reduction in conflicts among users, as also suggested by Boufous et al. (2018), who found that cyclists tend to ride faster on wider paths, a pattern confirmed in our study. Quantitatively, our results suggest that widening a 100-meter-long shared path used by 5,000 cyclists daily from 3.5 to 7 meters could result in approximately 2,700 cyclists reporting a lower perceived unsafety category. In addition, every cyclist would, on average, increase their speed on this stretch by 2.8 km/h.

At unsignalized intersections, our results show that cyclists feel significantly less safe when the speed of approaching vehicles is higher. This finding is consistent with Beura et al. (2021) and Jensen (2012), who suggest that higher vehicle speeds and increased traffic flow contribute to a reduced sense of safety among vulnerable road users. Moreover, we observe that cyclists' average speed decrease as they are forced to wait longer for acceptable gaps in traffic, leading to increased overall travel times. This highlights a

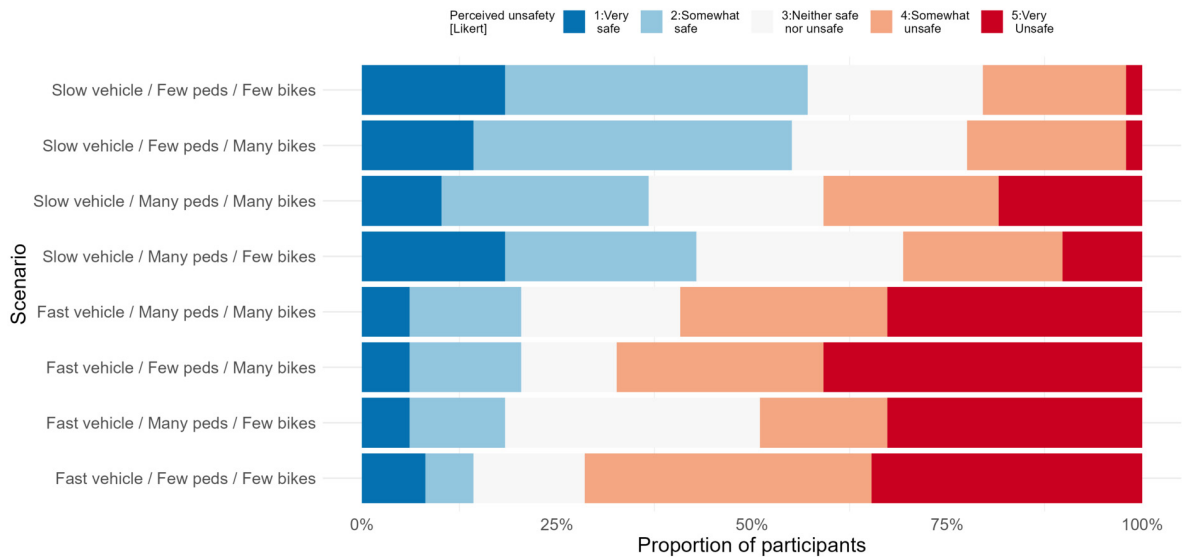


Fig. 8. Perceived unsafety in eight scenarios at the unsignalized intersection environment.

Table 7

Model estimation results for fixation on vehicles at the unsignalized intersection.

Variables	Coefficient	Std. error	t-value	p-value
Fixation on vehicles [%]				
Intercept	9.658	2.644	3.653	<0.001 ***
Participant average speed high	-0.262	0.083	-3.165	0.002 ***
Vehicle speed high	3.457	0.788	4.388	<0.001 ***
Pedestrian flow high	-2.176	0.703	-3.095	0.002 ***
Adjusted R-squared	0.578			

* (p-value <0.05), ** (p-value <0.01), and *** (p-value <0.005).

clear trade-off between car drivers' mobility and cyclists' mobility: while higher vehicle speeds may benefit drivers by reducing their travel times, they simultaneously reduce perceived safety and increase delays for cyclists. According to our results, at an unsignalized intersection with similar geometry to our experimental setup, reducing vehicle speed from 40 km/h to 20 km/h would lead to a 12 km/h increase in cyclists' average speed while approaching, crossing, and exiting the intersection. These findings underscore an important and often overlooked point in transport planning: cyclists' mobility is rarely integrated into traditional transport planning methods, which primarily focus on minimizing travel time for motorized vehicles and consider safety for active travelers. Our results emphasize the need for a more balanced approach that recognizes cyclists not merely as safety concerns but as legitimate road users with mobility needs. By explicitly incorporating cyclists' travel time and perceived safety into planning and modeling frameworks, policymakers and engineers can design intersections and networks that support equitable and sustainable transport systems.

In both environments, we found a similar tendency in the relationship between flows of other bicyclists and pedestrians and how this affects both the cyclists' perceived unsafety and their average speed. Generally, cyclists felt less safe as pedestrian and bicycle traffic increased at shared paths and unsignalized intersections. Previous studies for shared paths (Berghoefer & Vollrath, 2023; Gkekas et al., 2020) have found similar patterns. Interestingly, bicycle flow had nearly twice the impact on perceived unsafety than pedestrian flow at shared paths, likely because cyclists move faster and pose a greater risk of severe crashes. We also find that cyclists perceive a wider shared path with higher bicycle flow as safer than a narrower path with lower bicycle flow, even when both scenarios have the same traffic density (road users per area). This perception may stem from the broader visual space, which gives cyclists a sense of having more space to maneuver and reduces the perceived risk of collisions with other bicycles. Finally, we found a negative correlation between cyclists' perceived unsafety and participant average speed in both environments, largely due to common contributing factors affecting both indicators. This outcome is intuitive, as cyclists forced to ride slower tend to feel less safe. Improving cyclists' mobility can therefore enhance their perceived safety and vice versa.

These findings underscore the need for planners, architects, and engineers to go beyond observed safety, aesthetic, and experiential considerations when designing cycling infrastructure. By incorporating quantitative evidence on mobility and perceived safety

and behavior from studies like ours, they can better address users' actual needs and create environments that encourage cycling. Prioritizing wider paths can thus contribute to both improved subjective safety and more efficient flow, ultimately supporting broader policy goals of promoting active and sustainable transportation.

Finally, while our study highlights the importance of improving bicycle traffic flow and reducing perceived unsafety, these goals should never come at the expense of actual safety. Our results also provide new insights into how cyclists' attention to motorized traffic is shaped by various factors. Cyclists riding at higher speeds tend to focus less on vehicles, potentially prioritizing their own movement over monitoring traffic. In contrast, higher vehicle speeds increase cyclists' fixation on cars, reflecting a heightened need for caution. Interestingly, higher pedestrian flow appears to draw cyclists' attention away from vehicles, likely because they must navigate around other vulnerable road users. Determining the design of an intersection should thus depend on multiple contextual factors, including the dominant type of cyclists (e.g., commuters versus leisure riders), their typical speeds, and the level of pedestrian activity. By accounting for these elements and integrating both perceived and actual safety considerations, planners and engineers can better create intersections that accommodate diverse user needs.

5.1. Limitation and future research

Although we were able to quantify and confirm our hypotheses using the cycling simulator experiments, a few limitations are worth mentioning. In our setup, participants lack a 360-degree virtual environment for full immersion, and the static cycling simulator prevents them from leaning while steering. These technical limitations make the simulation less realistic. Furthermore, in the simulation environments, the visualized pedestrians and cyclists can only travel straight and cannot interact with participants. This may have led to behaviors and subjective safety assessments differing from those typically present in real life. At most unsignalized intersections in cities with high cycling contexts, vehicles often yield to cyclists, even when they have the right of way. Some participants expected this yielding behavior, which influenced their actions, despite being informed before the experiment that vehicles had priority and would not yield. Similar experiences are described in Berghoefer and Vollrath (2023).

Future research could enhance realism by improving the technical setups and user interactions in the simulation environment. Additionally, incorporating individual characteristics such as their riding styles, riding experience, and safety awareness could help explore how personal features influence cycling behaviors and subjective safety. Our experiment can also be replicated with different design configurations of shared paths and unsignalized intersections to gain more comprehensive insights into cyclists' behavior and perceived safety. Such research would help improve the understanding of cyclist safety, mobility, and riding experiences across these two types of infrastructure.

Finally, a few insights from our modeling process are worth highlighting. We explored the relationship between fixation on cars and cyclists' perceived unsafety in both the model for perceived unsafety and the model for fixation time. However, we did not find a plausible result that combined a clear direction and statistical significance in both models. While it would seem most intuitive that cyclists who feel less safe would fixate longer on cars, this effect was not significant in the model with fixation time as the dependent variable. Conversely, we did find a significant parameter for fixation time in the model describing perceived unsafety, indicating that cyclists who fixate longer on cars tend to report feeling more unsafe. This relationship is likely correlational rather than causal, and the overall model results do not provide a consistent or robust interpretation. Therefore, we chose not to include this relationship explicitly in our final models. Previous studies provide mixed evidence on this topic: Mohammadi et al. (2024) suggest that longer fixation on vehicles is associated with lower perceived safety because cyclists are uncertain about drivers' intentions to yield, making them feel more cautious. In contrast, Stülpnagel (2020) found that a higher perceived hazard can lead to shorter fixation times, as cyclists may shift their gaze to scan the environment more broadly.

6. Conclusion

This study uses a cycling simulator to examine and quantify how path widths on shared paths, vehicle speeds at unsignalized intersections, and pedestrian and bicycle flow in both environments affect cyclist behavior and subjective safety. The experiment employed a full factorial design, allowing participants to experience environments with all possible combinations of the three attributes. Behavioral and subjective safety data collected from participants were then analyzed using a statistical modeling approach. We found that participant average speed and perceived unsafety are significantly associated with all measured attributes and quantified each effect.

The study provides new evidence on how shared path width, vehicle speed at intersections, and surrounding traffic flows influence cyclists' perceived safety and riding behavior. Our findings confirm that wider paths significantly enhance cyclists' feelings of safety and promote higher cycling speeds, while higher vehicle speeds at unsignalized intersections reduce cyclists' sense of safety and slow their movement. These results emphasize the trade-off between motorized and non-motorized mobility and highlight the importance of designing infrastructure that prioritizes both perceived and actual safety for cyclists.

We also show that cyclists' perceptions are strongly affected by interactions with other road users. In particular, higher flows of pedestrians and cyclists generally increase perceived unsafety, yet a wider path with higher bicycle flow can still be perceived as safer due to increased maneuvering space. Furthermore, our analysis of gaze behavior suggests that cyclists' attention to vehicles depends on their own riding speed, vehicle speed, and pedestrian density, underlining the importance of considering user context when designing intersections and shared spaces.

Overall, our findings contribute to a growing body of evidence supporting the design of cycling environments that not only improve safety but also promote greater comfort and efficiency for cyclists. By integrating these insights into transport models and urban planning, cities can better support active travel and create more sustainable, user-centered transport systems.

CRedit authorship contribution statement

Kuan-Yeh Chou: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mads Paulsen:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Mette Møller:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Anders Fjendbo Jensen:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Data availability

The data that has been used is confidential.

Appendix A. Literature review

See Tables A.8 and A.9.

Table A.8

Summary of studies exploring cyclists' behavior and subjective safety at shared paths (Part 1).

Study	Study area	Objective	Data collection method	Attributes	Behavior Indicators	Subjective Indicators	Findings
Shared path							
Kazemzadeh et al. (2024)	Lund, Sweden	Explore cyclists' comfort during interactions in shared spaces, considering demographic heterogeneity	Intercept survey (594 cyclists)	1. Cyclist demographics (age/gender/education) 2. Cycling experience 3. Interaction type (passing/meeting)		Comfort	1. Female cyclists, older adults, and less experienced cyclists generally report lower comfort in shared spaces 2. Significant demographic differences affect comfort perception, highlighting the need for targeted infrastructure improvements.
von Stülpnagel and Rintelen (2024)	Berlin, Germany	Explore how environmental and demographic factors affect cyclists' perceived safety	Survey (21,500 participants)	1. Buffer type between cycling and pedestrian lane (None/Greens/Cobbled) 2. Sidewalk usage (Free/Cafes/shops)		Perceived safety	Cyclists feel safe on sidewalks shared with pedestrians, at the expense of pedestrians' subjective safety
Berghoefer and Vollrath (2023)	Braunschweig (Campus), Germany	Explore factors affecting cyclists' infrastructure preferences and subjective safety	Cycling simulator (39 participants)	Advisory lane/Cycle path/Footpath: 1. Pedestrian volume (5/25 [pedestrians]) 2. Vehicle volume (200/1000 [veh/h]) Street/Multi-use path: 1. Pedestrian volume (5/25 [pedestrians]) 2. Vehicle volume (100/400 [veh/h])		1. Mental Comfort 2. Interaction 3. Environment 4. Ease of Use 5. Physical Comfort	1. Cycle paths that are separated from other road users were most preferred, while crowded footpaths and uphill segments were least preferred 2. Cyclists riding on paths/lanes involving pedestrians felt less safe and comfortable when pedestrian volume was higher.
Batista et al. (2023)	Braunschweig, Germany	Explore factors affecting pedestrians' and cyclists' infrastructure preferences	Survey (408 participants)	1. Space configuration (central placement of design elements/segregation types) 2. Design elements (benches/fountains/bollards/play elements) 3. Presence of motor vehicles	1. Crossing path choice 2. Interaction behaviors	1. Comfort 2. Enjoyment level 3. Preference for design attributes	1. Both pedestrians and cyclists favor centrally located design elements and protective barriers. 2. Motor vehicles are perceived negatively by both pedestrians and cyclists. 3. Effective shared space designs incorporate integration, placemaking, and informal segregation.
Zhang et al. (2023)	Global (Systematic Review)	Explore interactions and safety between pedestrians and micro-mobility vehicles (MMVs) in shared spaces	Systematic literature review (87 publications)	1. Infrastructure features 2. Road user characteristics 3. Operation condition	1. Swerving 2. Passing 3. Overtaking 4. Lateral displacement	1. Perceived safety 2. Comfort level 3. Attitudes toward pedestrians	Key factors include sufficient infrastructure width, clear regulations, and speed management. Effective solutions include user education and infrastructure improvements.
Pashkevich et al. (2022)	Kraków, Poland	Explore visual attention and speeds of pedestrians, cyclists, and electric scooter (ES) riders	Field experiment with eye tracker (12 participants)	1. Visual attention distribution (roadway/roadside/all three road users) 2. Speeds and interaction behaviors	1. Fixation frequency and duration 2. Interaction behaviors (maneuvering)	1. Perceived safety 2. Comfort level	1. Riders (bicycles and ES) mainly focus visual attention on roadway and pedestrians, highlighting hazard awareness. 2. Pedestrians tend to look more at surroundings. 3. Cyclists and ES riders travel at similar speeds (16 km/h), suggesting ES should be regulated similarly to bicycles for safety.

Table A.8 (continued)

Study	Study area	Objective	Data collection method	Attributes	Behavior Indicators	Subjective Indicators	Findings
Che et al. (2021)	Singapore	Explore the relationship among perception, attitude, and behavior of cyclists and pedestrians	1. Survey for subjective indicators (300 cyclists) 2. Video observation for behavior indicator (Before keeping left rule: 119 cyclists After keeping left rule: 59 cyclists)	1. On-coming pedestrian-cyclist passing 2. Cyclist overtaking pedestrian 3. Rule of keep left for all road users	Lane-keeping action: 1. Keep left 2. Keep right 3. Zigzag	1. Perceived safety 2. Perceived control 3. Perceived priority	1. Cyclists' perceived control, perceived risk level, and perceived priority influence their attitude and further affect their behavior 2. Cyclists tend to compensate for pedestrians' risks, thereby indicating a more cooperative way of space-sharing
Huang et al. (2021)	Hefei (Campus), China	Explore how behavior of cyclists and different movements of pedestrian crowds affect each other	Closed-field experiment (95 participants)	1. Pedestrian and cyclist density (flow) 2. Pedestrian and cyclist flow direction (homogeneous/unidirectional/bidirectional)	1. Speed 2. Pedestrian lane formation		1. Cyclists travel slower when pedestrian density is higher 2. Pedestrian lanes were more concentrated in bidirectional flow than unidirectional flow
Gkekas et al. (2020)	Vancouver (Campus), Canada	Explore factors affecting the risk of pedestrian-cyclist interactions	Survey (337 participants)			1. Perceived safety 2. Comfort	1. Both pedestrians and cyclists considered crowding and pedestrian inattention as key factors contributing to incidents 2. Cyclists feel safer when paths get wider
Stülpnagel (2020)	Freiburg, Germany	Explore cyclists' gaze behavior relates to subjective risk perception and local spatial configurations	Field experiment with eye-tracking (20 participants)	1. Spatial complexity 2. Subjective risk perception	1. Sight vector length 2. Gaze angle 3. Fixation duration	1. Perceived safety	1. Cyclists' gaze behavior is strongly influenced by spatial openness and complexity; open areas increase perceived risk. 2. Experienced cyclists demonstrated longer sight vectors and narrower gaze angles, indicating more focused attention and hazard anticipation. 3. Recommendations for urban planning include balancing visibility and spatial structure to manage visual workload and enhance cyclists' perceived safety.

Table A.9

Summary of studies exploring cyclists' behavior and subjective safety at shared paths and unsignalized intersections (Part 2).

Study	Study area	Objective	Data collection method	Attributes	Behavior Indicators	Subjective Indicators	Findings
Nikiforiadis and Basbas (2019)	Thessaloniki, Greece	Explore factors influencing pedestrian perception of comfort and safety on shared sidewalks and pedestrian street	1. Survey (300 pedestrians) 2. Field measurements	1. Pedestrian and bicycle lane widths 2. Pedestrian and bicycle volumes and speeds		1. Comfort 2. Perceived safety 3. Space sufficiency	1. Pedestrian comfort and safety negatively impacted by increasing bicycle lane width relative to pedestrian area 2. Pedestrian perceived level of service strongly influenced by space sufficiency and pedestrian unit flow rate, emphasizing the importance of appropriate spatial design.
Beitel et al. (2018)	Montreal (Campus), Canada	Explore factors affecting the risk of pedestrian-cyclist interactions	1. Video observation for speed (2,200 cyclists) 2. Survey for conflict rate (872 responses)	Pedestrian density (flow)	1. Speed 2. Conflict rate 3. Angle of approach 4. Post-encroachment time		1. Cyclists travel slower when pedestrian density is higher 2. Conflict rate increase when pedestrian density is higher
Boufous et al. (2018)	Sydney, Australia	Explore how environmental factors affect cycling speed	Video observation (5,421 cyclists)	1. Width (≤ 3.5 / > 3.5 m) 2. Pedestrian flow (< 20 / 20 - 100 / 100 - 199 / ≥ 200) 3. Centerline (with/without) 4. Commuter path (with/without) 5. Visual segregation (yes/no)	Speed		Cyclists travel slower on shared paths with smaller widths, higher pedestrian flow, centerline, and visual segregation
Essa et al. (2018)	Vancouver, Canada	Explore factors affecting the risk of pedestrian-cyclist interactions	Video observation (940 cyclists, 800 streets)	1. Pedestrian density (flow) 2. Dismount sign	1. Speed 2. Conflict rate		1. Cyclists travel slower when pedestrian density is higher 2. Conflict rate decreases when bicycle dismount sign is presented
Unsignalized intersection							
Mohammadi et al. (2024)	Gothenburg, Sweden	Explore factors affecting cyclist behavior by interacting with vehicles	Cycling simulator (25 participants)	1. The difference in time to arrival at the intersection (DTA) (1.2/2.5/3.5 [s]) 2. Intersection visibility (IV) (22/27 [m]) 3. Pedaling 4. Looking duration	1. Average speed 2. Braking distance 3. Yielding decision		1. Cyclist's DTA and IV significantly affect braking distance 2. Cyclist's pedaling, looking duration, and DTA have a significant impact on cyclist's yielding behavior

Table A.9 (continued)

Study	Study area	Objective	Data collection method	Attributes	Behavior Indicators	Subjective Indicators	Findings
Mohammadi et al. (2023)	Gothenburg, Sweden	Explore factors affecting cyclist behavior by interacting with vehicles	Video observation (105 interactions)	1. Pedaling 2. Head movement 3. Hand gesture 4. Initial speed 5. The difference in time to arrival at the intersection (DTA)	Yielding decision		Cyclist's pedaling, head movement, DTA, and speed of cyclist and vehicle have a significant impact on cyclist's yielding behavior
Ng et al. (2017)	Queensland, Australia	Explore which types of cycling infrastructure cyclists perceive to be the safest at unsignalized intersections	Survey (214 Participants)	12 different types of cycling infrastructure with each type having three different interactions with vehicles		Perceived safety	1. Off-road bicycle paths and footpaths were perceived to be the safest cycling infrastructure at unsignalized intersections 2. Cyclists' perceived safety is influenced by motorists' yielding behavior toward them

Appendix B. Perceived unsafety and stress

See Figs. B.9 and B.10.

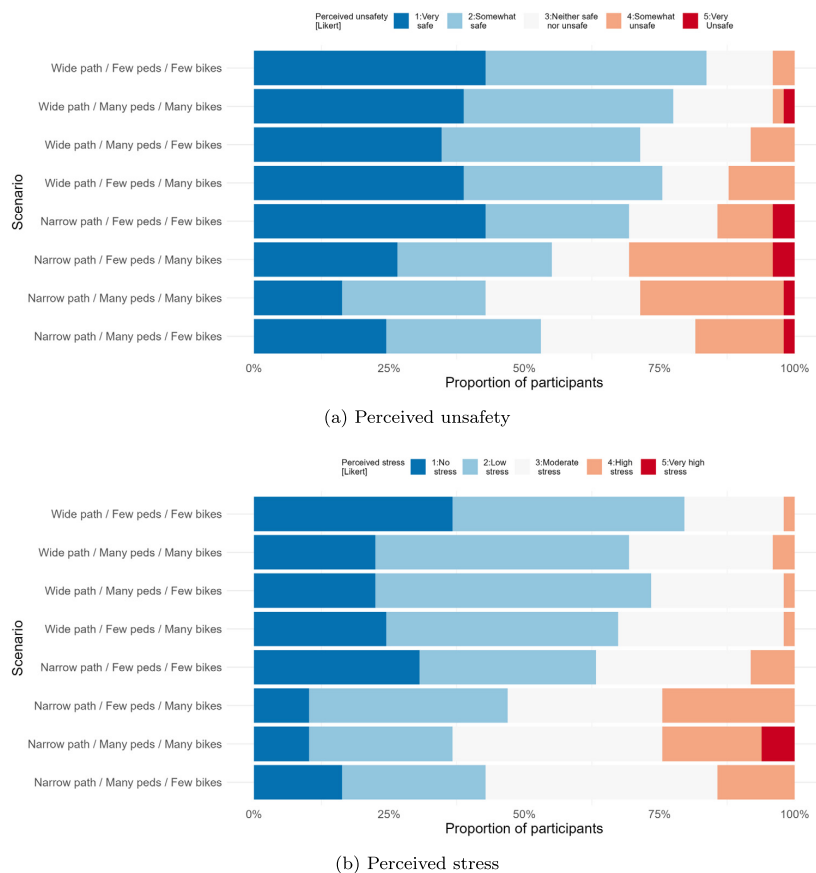


Fig. B.9. Perceived unsafety and stress in eight scenarios at the shared path environment.

Appendix C. Model estimation results with all variables

See Tables C.10–C.14.

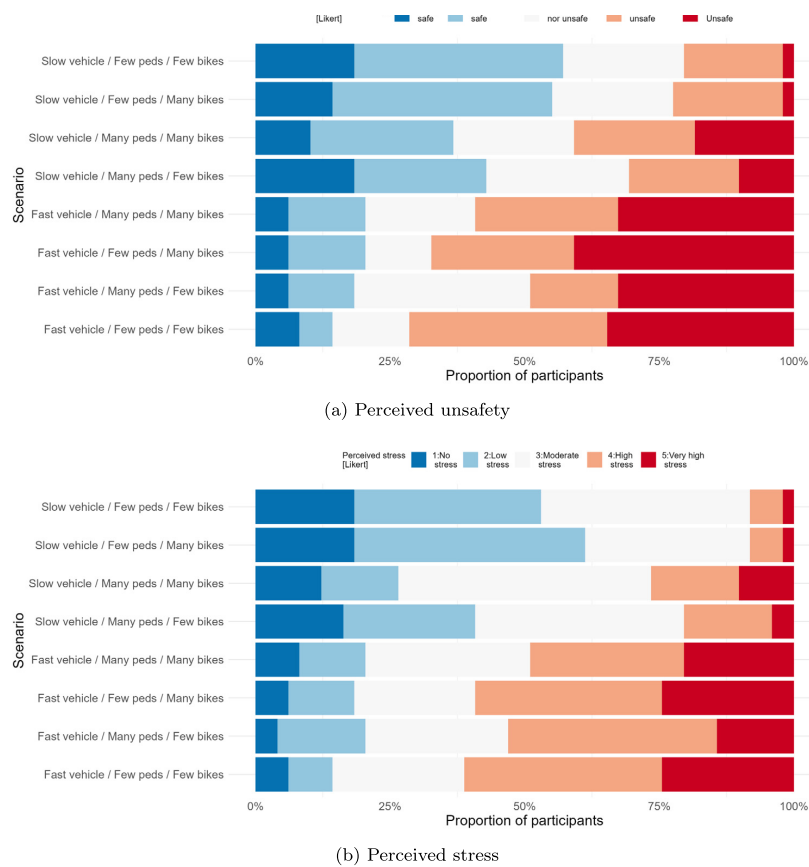


Fig. B.10. Perceived unsafety and stress in eight scenarios at the unsignalized intersection environment.

Table C.10

Model estimation results for participant average speed at the shared path.

Variables	Coefficient	Std. error	t-value	p-value
Participant average speed [km/hour]				
Intercept	17.001	1.206	14.094	<0.001 ***
Path width high	3.146	0.563	5.584	<0.001 ***
Pedestrian flow high	-1.907	0.563	-3.385	0.001 **
Bicycle flow high	-1.050	0.563	-1.864	0.063
Road width high $\times$ Pedestrian flow high	0.314	0.651	0.482	0.630
Path width high $\times$ Bicycle flow high	-0.941	0.651	-1.446	0.149
Pedestrian flow high $\times$ Bicycle flow high	1.870	0.651	2.874	0.004 **
Adjusted R-squared	0.485			

* (p-value <0.05), ** (p-value <0.01), and *** (p-value <0.005).

Table C.11

Model estimation results for perceived unsafety at the shared path.

Variables	Odds ratio	Std. error	t-value	p-value
Intercepts (Thresholds)		Perceived unsafety [Likert]		
Perceived unsafety 1 vs 2 (or higher)	0.018	1.119	-3.575	<0.001 ***
Perceived unsafety 2 (or lower) vs 3 (or higher)	0.355	1.097	-0.944	0.345
Perceived unsafety 3 (or lower) vs 4 (or higher)	3.033	1.101	1.088	0.313
Perceived unsafety 4 (or lower) vs 5	175.550	1.235	4.185	<0.001 ***
Participant average speed	0.892	0.041	-2.798	0.005 **
Road Width high	0.655	0.436	-0.972	0.331
Bicycle flow high	4.982	0.434	3.697	<0.001 ***
Pedestrian flow high	3.947	0.422	3.254	0.001 **
Fixation on pedestrians (%)	0.996	0.014	-0.274	0.784
Fixation on bicycles (%)	1.003	0.030	0.099	0.921
Road Width high × Pedestrian flow high	0.437	0.474	-1.749	0.080
Road Width high × Bicycle flow high	0.327	0.476	-2.374	0.018 *
Bicycle flow high × Pedestrian flow high	0.530	0.457	-1.335	0.182
Akaike information criterion (AIC)		682.75		

* (p-value <0.05), ** (p-value <0.01), and *** (p-value <0.005).

Table C.12

Model estimation results for participant average speed at the unsignalized intersection.

Variables	Coefficient	Std. error	t-value	p-value
Participant average speed [km/hour]				
Intercept	16.023	1.621	9.883	<0.001 ***
Vehicle speed high	-7.468	0.757	-9.862	<0.001 ***
Pedestrian flow high	-3.986	0.757	-5.263	<0.001 ***
Bicycle flow high	-0.754	0.757	-0.996	0.320
Vehicle speed high × Pedestrian flow high	4.547	0.874	5.200	<0.001 ***
Vehicle speed high × Bicycle flow high	1.579	0.874	1.806	0.072
Pedestrian flow high × Bicycle flow high	1.000	0.874	1.144	0.254
Adjusted R-squared		0.395		

* (p-value <0.05), ** (p-value <0.01), and *** (p-value <0.005).

Table C.13

Model estimation results for perceived unsafety at the unsignalized intersection.

Variables	Odds ratio	Std. error	t-value	p-value
Intercepts (Thresholds)		Perceived unsafety [Likert]		
Perceived unsafety 1 vs 2 (or higher)	0.072	0.887	-6.183	<0.001 ***
Perceived unsafety 2 (or lower) vs 3 (or higher)	0.072	0.828	-6.180	<0.001 ***
Perceived unsafety 3 (or lower) vs 4 (or higher)	0.540	0.818	-0.354	0.723
Perceived unsafety 4 (or lower) vs 5	4.059	0.822	1.704	0.088
Participant average speed high	0.795	0.033	-7.028	<0.001 ***
Vehicle speed high	5.468	0.438	7.883	<0.001 ***
Bicycle flow high	1.377	0.374	0.999	0.317
Pedestrian flow high	0.823	0.401	-0.389	0.757
Fixation on pedestrians (%)	1.020	0.014	1.369	0.171
Fixation on bicycles (%)	0.868	0.044	-1.495	0.135
Fixation on vehicles (%)	1.055	0.020	2.716	0.007 **
Vehicle speed high × Pedestrian flow high	0.600	0.458	-2.111	0.035 *
Vehicle speed high × Bicycle flow high	0.898	0.459	-0.223	0.823
Bicycle flow high × Pedestrian flow high	1.843	0.437	1.403	0.162
Akaike information criterion (AIC)		834.31		

* (p-value <0.05), ** (p-value <0.01), and *** (p-value <0.005).

Table C.14

Model estimation results for fixation on vehicles at the unsignalized intersection.

Variables	Coefficient	Std. error	t-value	p-value
Fixation on vehicles [%]				
Intercept	10.548	2.945	3.582	<0.001 ***
Participant average speed high	-0.238	0.087	-2.730	0.007 **
Perceived unsafety high	-0.556	0.546	-1.019	0.309
Vehicle speed high	4.257	1.382	3.080	0.002 **
Pedestrian flow high	-1.489	1.282	-1.162	0.246
Bicycle flow high	-0.363	1.215	-0.299	0.765
Vehicle speed high × Pedestrian flow high	-0.701	1.457	-0.481	0.631
Vehicle speed high × Bicycle flow high	-0.175	1.424	-0.123	0.902
Pedestrian flow high × Bicycle flow high	-0.465	1.419	-0.328	0.743
Adjusted R-squared	0.579			

* (p-value <0.05), ** (p-value <0.01), and *** (p-value <0.005).

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